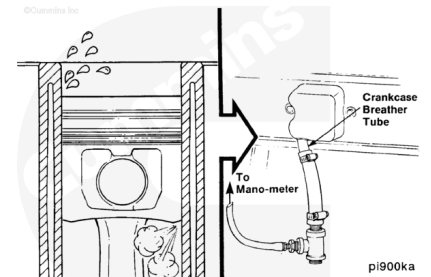


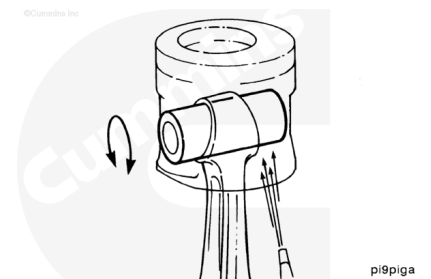
Trainings ▾

General Information

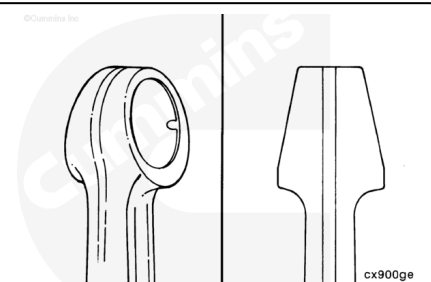
There are a number of power-related problems, including excessive lubricating oil consumption, smoke, blowby, and poor performance that can be caused by inadequate sealing between the piston rings and the cylinder walls. A blowby measurement can help to detect these problems.



A free-floating, hollow piston pin is used to attach the piston to the connecting rod. Lubrication of the pin and journal is accomplished by residual spray from piston cooling.



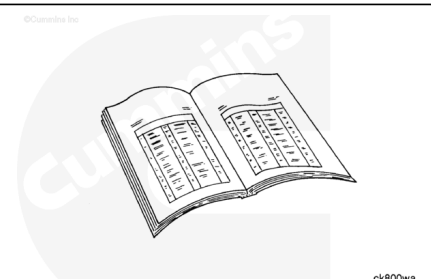
The piston pin end of the connecting rod is angle-cut to provide additional bearing surface. The connecting rod end is fitted with a bronze bushing.



Preparatory Steps

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. If not reused, dispose of in accordance with local environmental regulations.

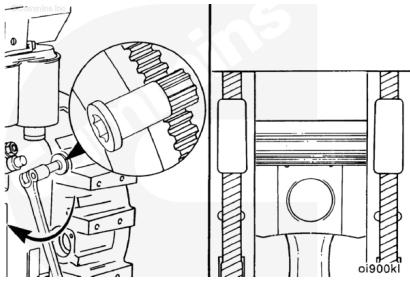
⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).
- Drain the coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>).
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2 (</qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html>).
- Mark each connecting rod cap according to cylinder.

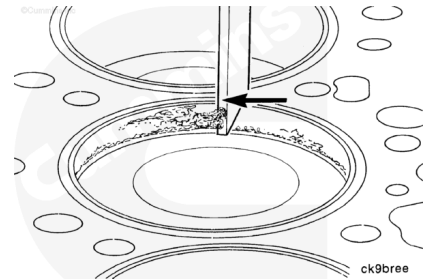
Remove

Rotate the crankshaft with engine barring tool, Part Number 3377371, until the pistons are below the carbon deposits that are found above the ring travel area.



⚠ CAUTION ⚠

Do not use emery cloth or sandpaper to remove carbon from the cylinder liners. Aluminum oxide or silicon particles from emery cloth or sandpaper can cause serious engine damage. Do not use any abrasives in the ring travel area. The cylinder liner can be damaged.

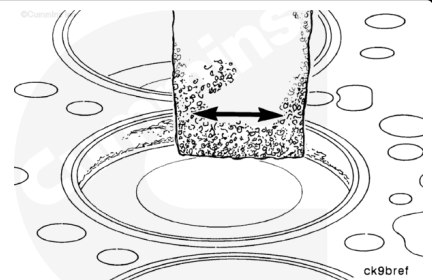


Use a scraper or a blunt-edged instrument to loosen the carbon deposits. Do **not** damage the cylinder liner with the scraper.

Note : Only scrape or sand above the piston travel area.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



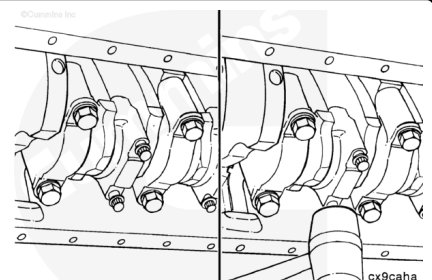
Remove the remaining carbon with Scotch-Brite™ 7448 abrasive pad, Part Number 3823258, and solvent.

Rotate the crankshaft to position the connecting rod caps just before Top Dead Center for removal.

Note : Do not remove the capscrews from the connecting rods.

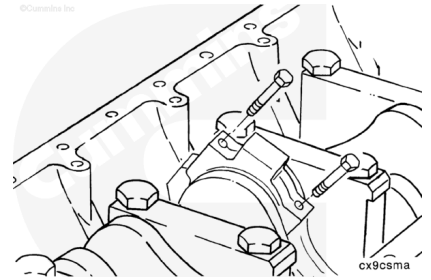
Loosen the connecting rod capscrew nuts.

Use a plastic hammer to hit the connecting rod capscrews to loosen the connecting rod caps.



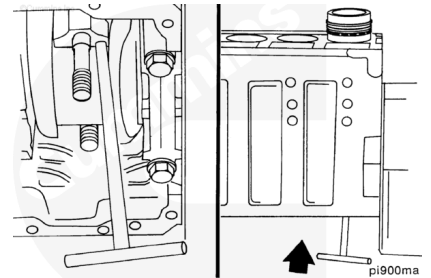
Remove the capscrew nuts, connecting rod cap, and connecting rod bearings.

Mark the connecting rod bearings with the cylinder number and "U" for the upper bearing and "L" for the lower bearings.



Use the T-handle piston pusher to push the rod away from the crankshaft.

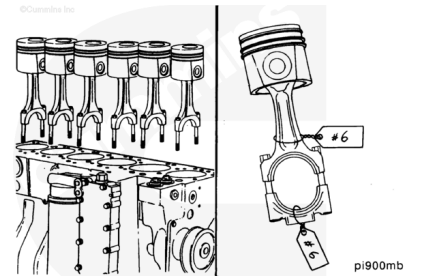
Note : Push the rod until the piston rings are outside the top of the cylinder liner.



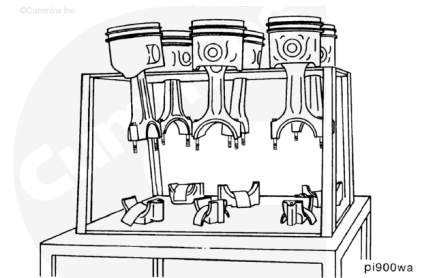
Use both hands to remove the piston and rod assembly.

Note : The piston and connecting rod assemblies must be installed in the same cylinder from which they were removed. This will allow for the proper fit of worn mating surfaces if the parts are used again.

Use a tag to mark the cylinder number from which each piston and rod assembly was removed.

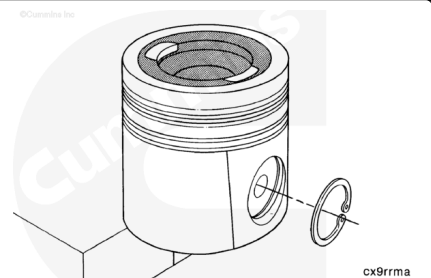


Place the connecting rod and piston assemblies in a container to protect them from damage.



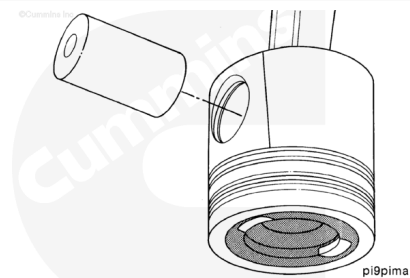
Disassemble

Remove the piston pin retaining rings.

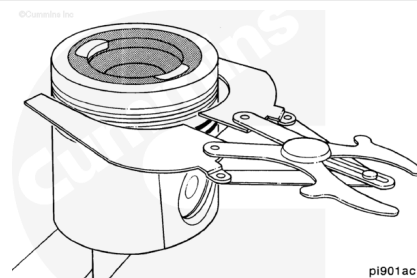


Note : Heating the piston is **not** required.

Remove the piston pin.

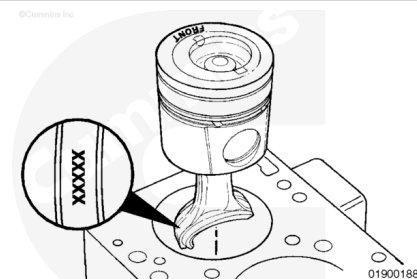


Remove the piston rings with piston ring expander, Part Number 3823137.

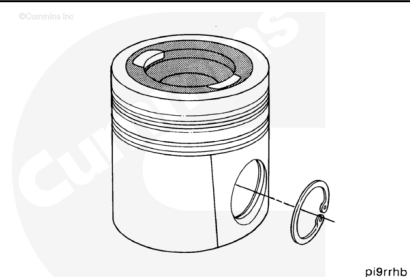


Assemble

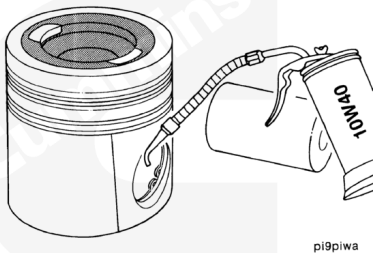
Make sure FRONT marking on the piston and the numbers on the connecting rod and cap are oriented as illustrated.



Install the retaining ring in the piston pin groove on the front side of the piston.

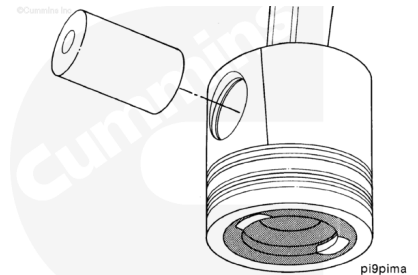


Lubricate the piston pin and piston pin bores with clean engine oil.

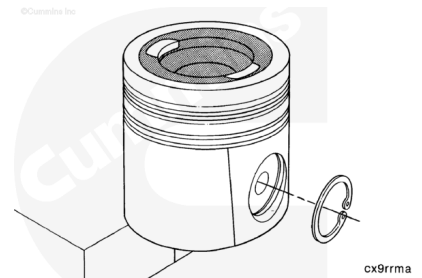


Note : Pistons do not require heating to install the pin; however, the pistons do need to be at room temperature or above.

Install the piston pin.



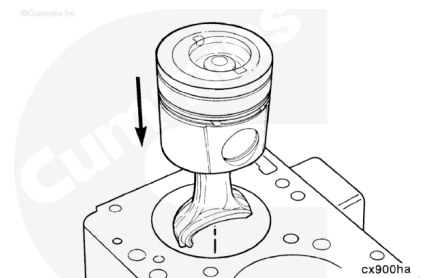
Install the second piston pin retaining ring.



Piston Grading

When rebuilding an engine with the original cylinder block, crankshaft, and pistons, make sure the pistons are installed in the original cylinder. If replacing the piston(s), make sure the replacement piston(s) is the same grade as the original piston. If a new cylinder block or crankshaft is used, the piston grading procedure **must** be performed to determine the proper piston grade for each cylinder.

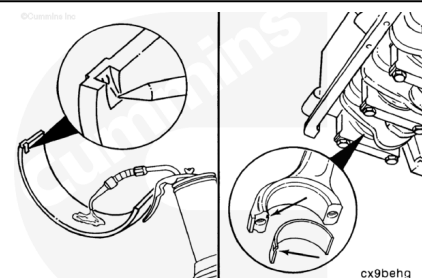
Install the connecting rod/piston assembly into the Number 1 cylinder without the rings installed.



Note : The connecting rod bearings must be installed in the original connecting rod and cap.

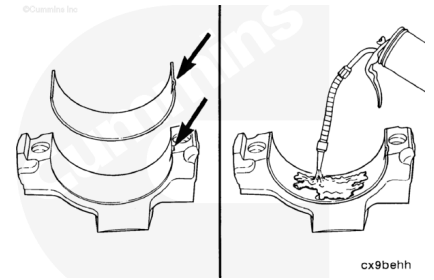
Install the upper bearing shell in the connecting rod with the tang of the bearing in the slot of the connecting rod.

Use clean lubricating oil to coat the inside of the connecting rod bearing shell.



Install the bearing shell in the connecting rod cap with the tang of the bearing in the slot of the cap.

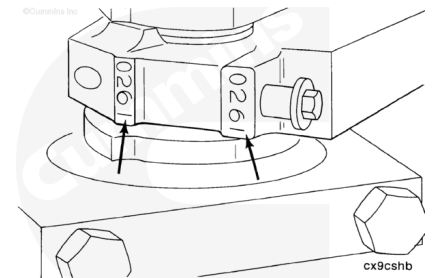
Use clean lubricating oil to coat the inside diameter of the connecting rod bearing shell.



The four-digit number stamped on the connecting rod and cap at the parting line **must** match and be installed on the same side in all cylinders.

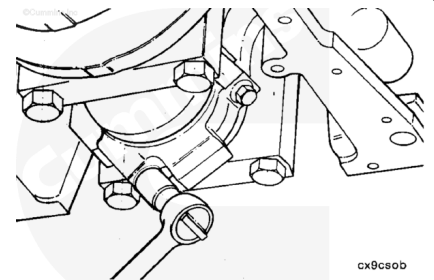
Install the connecting rod cap and capscrews to the connecting rod.

Note : The word **FRONT** on the piston will now be toward the front of the block. The combustion bowl in the piston will be on the fuel pump side of the engine.



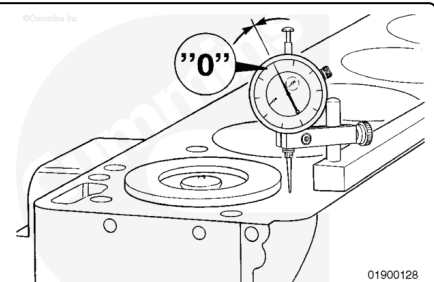
Tighten the two connecting rod capscrews.

Torque Value: 40 n·m [30 ft-lb]



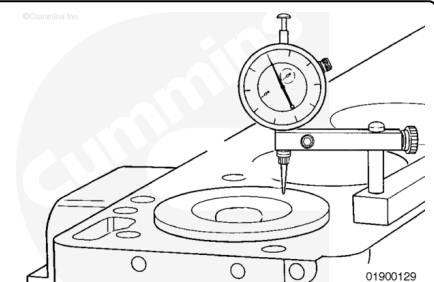
Use a fine-grit hone to remove any burrs from the cylinder block head deck.

Zero the dial indicator, Part Number 3823495, to the cylinder block head deck.



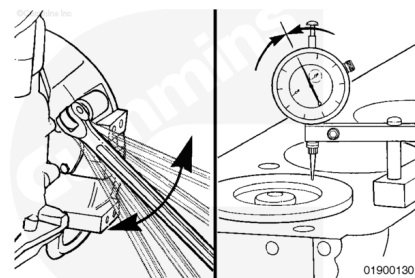
Move the dial indicator over the piston directly over the piston pin to eliminate any side-to-side movement.

Do **not** place the indicator tip on the anodized area.

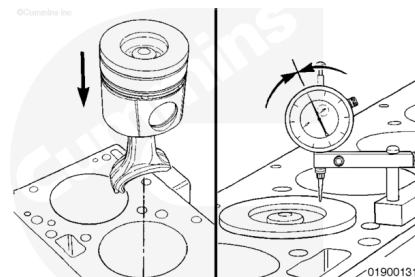


Rotate the crankshaft to top dead center. Rotate the crankshaft **clockwise** and **counterclockwise** to find the highest dial indicator reading.

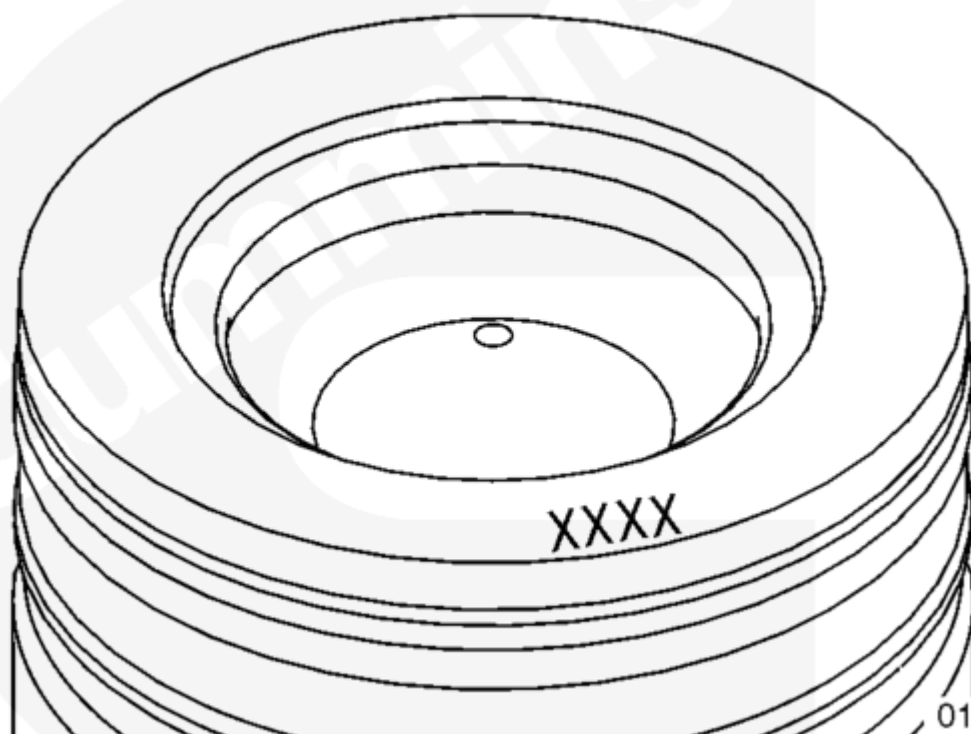
Record the reading.



Remove the piston connecting rod assembly from the Number 1 cylinder, and install the assembly into the Number 2 cylinder. Repeat the procedure for every cylinder using the same piston connecting rod assembly.



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Determine the grade of the piston being used by referring to the chart.

The four digits on top of the piston are the last four digits of the part number.

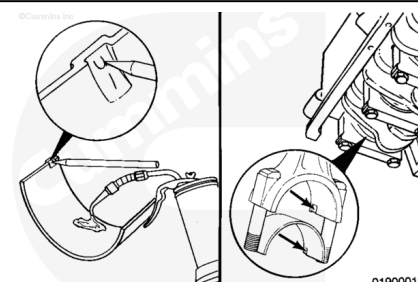
Piston Protrusion				
Measuring Piston	Measured Protrusion	Use Grade	210 to 250 hp	275 to 300 hp
A	0.58 to 0.65 mm [0.022 to 0.025 in]	A	3925878	3926961
A	0.51 to 0.58 mm [0.020 to 0.022 in]	B	3925879	3926962
A	0.44 to 0.51 mm [0.017 to 0.020 in]	C	3925880	3926963
A	0.37 to 0.44 mm [0.014 to 0.017 in]	D	3925881	3926964
B	0.65 to 0.72 mm [0.025 to 0.028 in]	A	3925878	3926961
B	0.58 to 0.65 mm [0.022 to 0.025 in]	B	3925879	3926962
B	0.51 to 0.58 mm [0.020 to 0.022 in]	C	3925880	3926963
B	0.44 to 0.51 mm [0.017 to 0.020 in]	D	3925881	3926964
C	0.72 to 0.79 mm [0.028 to 0.031 in]	A	3925878	3926961
C	0.65 to 0.72 mm [0.025 to 0.028 in]	B	3925879	3926962
C	0.58 to 0.65 mm [0.022 to 0.025 in]	C	3925880	3926963
C	0.51 to 0.58 mm [0.020 to 0.022 in]	D	3925881	3926964

Piston Protrusion				
Measuring Piston	Measured Protrusion	Use Grade	210 to 250 hp	275 to 300 hp
D	0.79 to 0.86 mm [0.031 to 0.033 in]	A	3925878	3926961
D	0.72 to 0.79 mm [0.028 to 0.031 in]	B	3925879	3926962
D	0.65 to 0.72 mm [0.025 to 0.028 in]	C	3925880	3926963
D	0.58 to 0.65 mm [0.022 to 0.025 in]	D	3925881	3926964

Install

Piston Protrusion for Emission-Controlled Engines

	mm		in
Engines built after 1/1/1994	0.584	MIN	0.023
	0.635	MAX	0.025

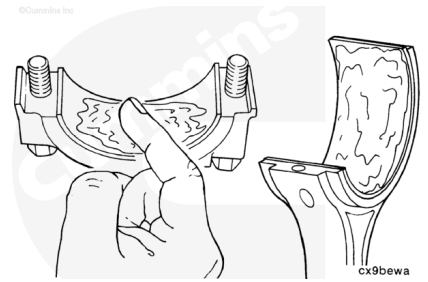


Install the bearing shells into both the connecting rod and the connecting rod cap. Make sure the tang on the bearing shell is in the slot of the connecting rod cap and connecting rod.

Note : If used connecting rod bearing shells are to be installed, each bearing shell must be installed in its original location.

⚠ CAUTION ⚠

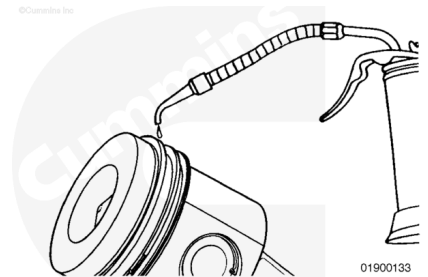
Do not lubricate the back of the connecting rod bearing. Prevent dirt from mixing with the lubricant. Dirty lubricant will accelerate connecting rod bearing wear.



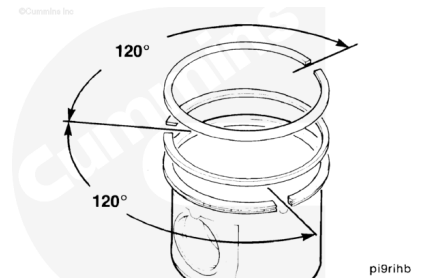
Lubricate the connecting rod bearings with a thin film of clean engine oil.

Install the piston rings. Refer to Procedure 001-047 in Section 1 (/qs3/pubsys2/xml/en/procedures/100/100-001-047.html).

Lubricate the rings and piston skirts with clean engine oil.



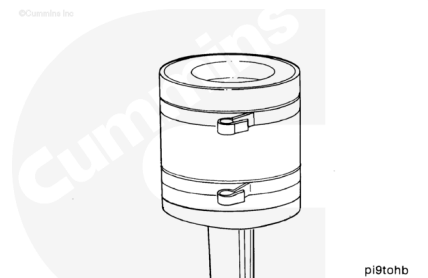
Make sure the piston ring end gaps are positioned 120 degrees apart.



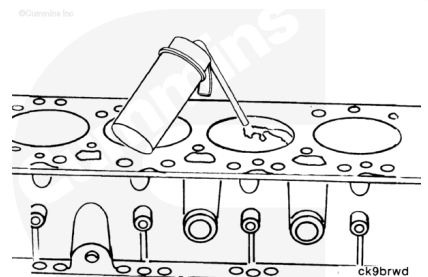
⚠ CAUTION ⚠

If using a strap-type ring compressor, make sure the inside end of the strap does not hook on a ring gap and break the ring.

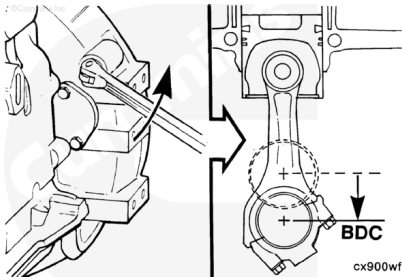
Compress the rings.



Lubricate the cylinder bore with clean engine oil.

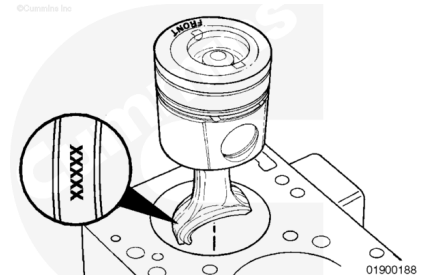


Position the connecting rod journal for the piston to be installed to Bottom Dead Center.

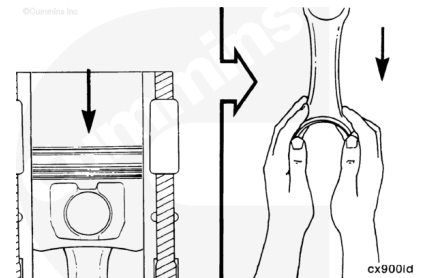


Take care **not** to damage the cylinder wall when inserting the connecting rod.

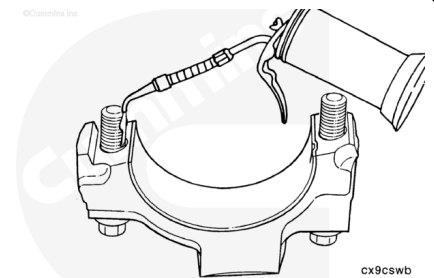
Note : The word **FRONT** stamped on the piston crown will be toward the front of the block. The piston bowl will then be on the fuel pump side of the engine.



Carefully push the piston into the bore while guiding the connecting rod to the crankshaft journal.



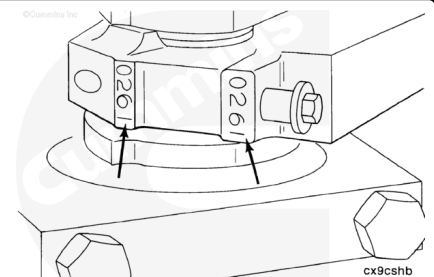
Lubricate the threads and underside of the connecting rod capscrew heads with clean engine oil.



⚠ CAUTION ⚠

The number stamped on the rod and cap at the parting line must match and be installed on the oil cooler side of the engine.

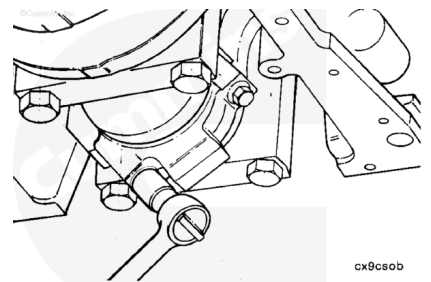
Install the connecting rod cap and capscrews to the connecting rod.



Alternately, tighten the two connecting rod capscrews.

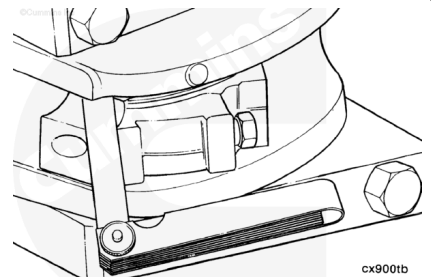
Torque Value:

1. 30 n•m [22 ft-lb]
2. 70 n•m [52 ft-lb]
3. 90 degree turn



Note : Do not measure the clearance between the rod cap and crankshaft.

Measure the side clearance between the connecting rod and crankshaft.



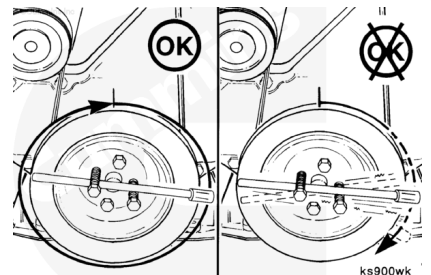
Side Clearance Limits

mm		in
0.10	MIN	0.004
0.33	MAX	0.013

Note : The crankshaft must rotate freely.

Check for freedom of rotation as the connecting rod caps are installed. If the crankshaft does **not** rotate freely, check the installation of the connecting rod bearings and the bearing size.

Note : If the connecting rod is not properly oriented (tang opposite the camshaft), it will contact the camshaft and lock the engine.



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



- Install the cylinder head. Refer to Procedure 002-004 in Section 2 (/qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html).

- Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7 (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>).
- Fill the engine with coolant. Refer to Procedure 008-018 in Section 8 (</qs3/pubsys2/xml/en/procedures/100/100-008-018.html>).
- Fill the engine with clean lubricating oil. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>).

Operate the engine and check for leaks.